4/12/16 Discussion section notes

$$\vec{D}(\vec{r},t) = \varepsilon \vec{E}(\vec{r},t)$$

$$\vec{D}(\vec{r},t) = \int d\vec{r}' \int_{-\infty}^{t} dt' \, \varepsilon(\vec{r}-\vec{r}', t-t') \vec{E}(\vec{r},t)$$

$$\vec{D}(\vec{r},t) \rightarrow \vec{D}(\vec{k},\omega)$$

Fourier transform

$$\vec{D}(\vec{k},\omega) = \varepsilon(\vec{k},\omega)\vec{E}(\vec{k},\omega)$$

$$\left.\frac{d}{dt} \rightarrow -i\omega \qquad \nabla \rightarrow i\vec{k}\right\}$$ replacements in Maxwell's Equations

Drude model: $\varepsilon(\omega) = 1 - \frac{\omega_0^2}{\omega^2}$, $\omega \gg vk$

Thomas Fermi: $\varepsilon(k) = 1 + \frac{\kappa^2}{k^2}$, $\omega \ll vk$

Electron-ion plasma, $m_e \ll M$

Jellium model:

$$\left.\varepsilon(\omega,k) = 1 - \frac{\omega_i^2}{\omega^2} + \frac{\kappa_e^2}{k^2}\right\}$$ ω_i = ion plasma frequency, $v_i k \ll \omega \ll v_e k$

- longitudinal waves $\rightarrow \vec{E} \parallel \vec{k}$
- find $\omega(\vec{k})$

Maxwell's Equations:

$$\nabla \cdot \vec{D} = 0 \rightarrow \nabla \cdot (\varepsilon(\omega,k)\vec{E}) = 0$$

$$\nabla \cdot \vec{B} = 0$$

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\nabla \times \vec{H} = \frac{\partial \vec{D}}{\partial t} = \frac{\partial}{\partial t}\left(\varepsilon(\omega,k)\vec{E}\right)$$

$$\left[\vec{\nabla}\varepsilon(\omega,k)\right]\vec{E} + \varepsilon(\omega,k)\,\vec{k}\cdot\vec{E} = 0$$

$$i\vec{k}\,\varepsilon(\omega,k)\vec{E} + \varepsilon(\omega,k)\,\vec{k}\cdot\vec{E} = 0$$

$$i\vec{k}\times\vec{H} = -i\omega\,\varepsilon(\omega,k)\vec{E}$$

$$\left.\begin{aligned} i\vec{k}\times\vec{E} &= i\omega\vec{B} \\ \vec{B} &= \mu_0\vec{H} \end{aligned}\right\} \quad \vec{k}\times\vec{E} = 0 \;\therefore\; \vec{B} = 0 \quad \therefore\; \vec{H} = 0$$

$$i\vec{k}\times\vec{H} = 0 = -i\omega\,\varepsilon(\omega,k)\vec{E}$$

if $\vec{E}$ is non-zero, $\varepsilon(\omega,k)$ must be zero.

$$0 = 1 - \frac{\omega_i^2}{\omega^2} + \frac{\kappa_e^2}{k^2} \longrightarrow \frac{\omega_i^2}{\omega^2} = 1 + \frac{\kappa_e^2}{k^2}$$

$$\boxed{\omega = \sqrt{\frac{\omega_i^2}{1 + \kappa_e^2/k^2}}}$$

When wavelength is long, $\kappa \gg k$, $\omega(k) \approx \omega_i$

$$\omega_p^2 = \frac{ne^2}{\varepsilon_0 m_e} \qquad \omega_i^2 = \frac{ne^2}{\varepsilon_0 m_i} \qquad v_i k \ll \omega \ll v_e k$$

$$\kappa_e^2 = \underbrace{\frac{e^2\nu_F}{\varepsilon_0}, \quad \nu_F \sim \frac{n}{E_F}}_{\kappa_e^2 = \frac{e^2 n}{\varepsilon_0 E_F}}$$

$$\omega = \sqrt{\frac{E_F}{m_i}}\,k$$

$$u = v_e\sqrt{\frac{m_e}{m_i}}$$ } velocity of ions is slow compared to electrons, same order of magnitude as sound waves in metal